NEXT ENGINEER · _SHEERAT_

Fundamentals of Electronics Engineering

AEC101A

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** CSE, IT, CSE-AIML, CSE-DS, AIDS, ECE, EE | **Semester:** 1 | **Credits:** 3

L-T-P: 3-0-0 | **Scheme:** AKTU B.Tech 1st year, 2026-27 onwards

UNIT 1 Diodes and Power Supplies

UNIT 2 Transistors (BJT and MOSFET)

UNIT 3 Operational Amplifiers

UNIT 4 Digital Electronics

UNIT 5 Electronic Systems and Interfacing

Original revision notes written for the AKTU syllabus. The same course is taught as AEC201A in semester 2 for the other group of students. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: circuits described in words, formulas and comparison tables.
3. **Solved examples**: the numericals examiners set (rectifier output, ripple, Zener regulator, LED resistor, BJT currents and bias point, transistor switch, MOSFET current, op-amp gains, number systems, K-map, ADC resolution, PWM, UART, LDO).
4. **Key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (pn junction, rectifiers, filter, Zener regulator, LED) and Unit 2 (BJT regions, CE amplifier, transistor switch, MOSFET). Practise the Zener and bias-point numericals.
- **Afternoon (3 hours):** Unit 3 (ideal op-amp, inverting, non-inverting, summing, comparator) and Unit 4 (number systems, Boolean algebra, K-map, adders, MUX, flip-flops).
- **Evening (2 hours):** Unit 5 (microcontroller, ADC, PWM, UART, SPI, I2C, wireless, power management, IoT examples), then the "Last-hour checklist".
- **In the exam:** draw the diode V-I curve, the bridge rectifier with its waveforms, the Zener regulator, the CE amplifier, the transistor switch with relay and flyback diode, each op-amp circuit, the K-map with groups circled, and a block diagram for every system question.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit. Internal assessment carries another 30 marks.

UNIT 1 Diodes and Power Supplies

✍️ 2-mark definitions

- **pn junction:** the boundary formed when p-type and n-type regions are made in one crystal. Diffusion of carriers leaves a **depletion region** of fixed ions with a built-in barrier potential (about 0.7 V for silicon, 0.3 V for germanium).
- **Forward bias:** p side made positive with respect to n side. The barrier falls and a large current flows once the voltage passes the **cut-in (knee) voltage**.
- **Reverse bias:** p side negative. The depletion region widens and only a tiny reverse saturation current (nA to µA) flows until **breakdown**.
- **Ideal diode:** a perfect switch: zero resistance when forward biased, infinite resistance when reverse biased.
- **Rectifier:** a circuit that converts AC to pulsating DC using diodes.
- **Ripple factor (γ):** the ratio of the RMS value of the AC component to the DC value of the output. Smaller is better.
- **Peak inverse voltage (PIV):** the largest reverse voltage a diode must withstand in a circuit. The chosen diode's rating must exceed it.
- **Zener diode:** a heavily doped diode designed to work in reverse breakdown, where its voltage stays almost constant (V_Z) over a wide range of current. Used as a voltage regulator.
- **LED:** a pn junction that emits light when forward biased, as electrons recombine with holes (GaAs, GaAsP, GaN). Forward drop is about 1.8 V to 3.3 V depending on colour.
- **Line and load regulation:** how much the output voltage changes with a change in input voltage (line) or load current (load).

💡 Core concepts

Diode models (approximations):

1. **Ideal:** short when on, open when off.
2. **Constant-drop (practical):** a 0.7 V drop when on, open when off. Use this in numericals unless told otherwise.
3. **Piecewise-linear:** 0.7 V source in series with a small bulk resistance r_d. - **V-I characteristic:** forward current rises exponentially after cut-in; reverse current stays tiny until breakdown (avalanche or Zener). The diode equation is I = I_S(e^(V/ηV_T) − 1), with V_T ≈ 26 mV at room temperature. - **Diode as a switch:** on when forward biased, off when reverse biased. Used for reverse-polarity protection and as the flyback diode across a relay coil.

Rectifiers (V_m = peak secondary voltage):

Quantity	Half-wave	Full-wave centre-tap	Full-wave bridge
Diodes	1	2	4
V_dc	V_m/π	2V_m/π	2V_m/π

Quantity	Half-wave	Full-wave centre-tap	Full-wave bridge
V_rms	V_m/2	V_m/√2	V_m/√2
Ripple factor	1.21	0.482	0.482
Efficiency (max)	40.6%	81.2%	81.2%
Ripple frequency	f (50 Hz)	2f (100 Hz)	2f (100 Hz)
PIV per diode	V_m	2V_m	V_m
Transformer	Ordinary	Centre-tapped needed	Ordinary

- **Bridge working:** in the positive half cycle D1 and D2 conduct; in the negative half cycle D3 and D4 conduct. Current through the load is in the same direction in both halves.
- **Bridge advantages:** no centre-tap, PIV is only V_m, smaller transformer. Disadvantage: two diode drops (1.4 V) in the path.

Capacitor filter:

- A large capacitor across the load charges to the peak and discharges slowly through the load between peaks.
- **Peak-to-peak ripple:** V_r ≈ I_dc/(f_r C), where f_r is the ripple frequency (2f for full wave). So V_r = I/(2fC) for a full-wave rectifier.
- Bigger C or smaller load current gives less ripple.

Zener voltage regulator:

- A series resistor R_S from the unregulated supply, with the Zener and load R_L in parallel.
- I_S = (V_in − V_Z)/R_S; I_L = V_Z/R_L; I_Z = I_S − I_L.
- The Zener absorbs the extra current to keep V_Z constant. It must stay above I_Z(min) and below I_Z(max) = P_Z/V_Z.

LED with series resistor: R = (V_supply − V_F)/I_F. Without it, the current would rise until the LED burns out.

Applications:

- **5 V / 3.3 V DC supply chain:** transformer (step down) → bridge rectifier → capacitor filter → regulator IC (7805 for 5 V, an LDO such as AMS1117 for 3.3 V) → load.
- **USB charger:** a mains adapter that rectifies 230 V AC, uses a switching (flyback) converter with a high-frequency transformer for isolation, and gives a regulated 5 V DC on the USB port.

🧮 Solved examples

Example 1 (rectifier output). A transformer gives a peak secondary voltage of 12 V. Find V_dc for a half-wave and a bridge rectifier (ideal diodes).

- Half-wave: V_dc = V_m/π = 12/π = **3.82 V**.
- Bridge: V_dc = 2V_m/π = 24/π = **7.64 V**. (With practical diodes the bridge peak falls to 12 − 1.4 = 10.6 V.)

Example 2 (filter ripple). A full-wave rectifier on 50 Hz mains feeds 100 mA into a load with a 1000 µF filter capacitor. Find the peak-to-peak ripple.

- $V_r = I/(2fC) = 0.1/(2 \times 50 \times 1000 \times 10^{-6}) =$ **1 V**.

Example 3 (Zener regulator). $V_{in} = 12$ V, $V_Z = 5.1$ V, $R_S = 220\ \Omega$, $R_L = 510\ \Omega$. Find I_S, I_L, I_Z and the Zener power.

- $I_S = (12 - 5.1)/220 =$ **31.4 mA**.
- $I_L = 5.1/510 =$ **10 mA**.
- $I_Z = 31.4 - 10 =$ **21.4 mA**.
- $P_Z = 5.1 \times 0.0214 =$ **0.109 W**, so a 0.5 W Zener is safe.

Example 4 (LED resistor). A red LED ($V_F = 2$ V) must carry 10 mA from a 5 V supply. $R = (5 - 2)/0.01 =$ **300 Ω**; use the next standard value, 330 Ω.

Key points to remember

- Silicon cut-in 0.7 V; $V_T \approx 26$ mV; reverse current tiny until breakdown.
- Half-wave: V_m/π, $\gamma = 1.21$, $\eta = 40.6\%$. Full-wave: $2V_m/\pi$, $\gamma = 0.482$, $\eta = 81.2\%$.
- PIV: half-wave V_m, centre-tap $2V_m$, bridge V_m.
- Filter ripple $V_r = I/(2fC)$; Zener: $I_Z = I_S - I_L$.
- LED resistor $R = (V - V_F)/I$.

⭐ Important questions

1. Explain the pn junction under forward and reverse bias with the V-I characteristic. Compare the ideal and practical diode.
2. Explain the full-wave bridge rectifier with circuit and waveforms. Derive V_{dc}, ripple factor and efficiency.
3. Explain the capacitor filter and derive the ripple voltage.
4. Explain the Zener diode as a voltage regulator and solve a numerical.
5. Draw the block diagram of a regulated 5 V / 3.3 V power supply and explain each block. How does a USB charger work?

UNIT 2 Transistors (BJT and MOSFET)

✍️ 2-mark definitions

- **BJT (bipolar junction transistor):** a three-layer device (emitter, base, collector) in NPN or PNP form, controlled by base current. Both electrons and holes carry current.
- **Emitter, base, collector:** the emitter is heavily doped (supplies carriers), the base is thin and lightly doped, and the collector is moderately doped and largest (collects carriers and dissipates heat).
- **Current relations:** $I_E = I_B + I_C$; $\beta = I_C/I_B$ (typically 50 to 300); $\alpha = I_C/I_E = \beta/(\beta + 1)$, just under 1.
- **Regions of operation: cut-off** (both junctions reverse biased, transistor off), **active** (BE forward, BC reverse, used for amplification), **saturation** (both forward biased, transistor fully on, $V_{CE(sat)} \approx 0.2$ V).
- **Q-point (operating point):** the DC values (I_C, V_{CE}) set by biasing, placed in the middle of the active region for an amplifier.
- **FET:** a unipolar, voltage-controlled transistor; the gate voltage controls current in a channel between source and drain. It has very high input resistance.
- **MOSFET:** a FET whose gate is insulated from the channel by a thin oxide layer. An **n-channel enhancement MOSFET** conducts only when V_{GS} exceeds the threshold voltage V_T.
- **PWM (pulse width modulation):** switching a load fully on and off quickly; the average power is controlled by the **duty cycle** $D = t_{on}/T$.

💡 Core concepts

NPN working (active region): the forward-biased BE junction injects electrons from the emitter into the thin base. Very few recombine (small I_B); most diffuse across into the reverse-biased collector, so $I_C \approx \beta I_B$.

CE (common-emitter) amplifier:

- Input between base and emitter, output between collector and emitter.
- Voltage gain $A_v \approx -g_m R_C$ (or $-\beta R_C/r_{in}$); the output is **180° out of phase** with the input.
- High voltage and current gain, so CE is the most used configuration.
- **Fixed bias:** $I_B = (V_{CC} - V_{BE})/R_B$; $I_C = \beta I_B$; $V_{CE} = V_{CC} - I_C R_C$.
- Voltage-divider bias with an emitter resistor gives a more stable Q-point, because it is not sensitive to β.

Comparison of configurations:

Feature	CB	CE	CC (emitter follower)
Input resistance	Low	Medium	High
Output resistance	High	Medium	Low

Feature	CB	CE	CC (emitter follower)
Current gain	α (< 1)	β (high)	β + 1
Voltage gain	High	High	≈ 1
Phase shift	0°	180°	0°
Use	High-frequency	General amplifier	Buffer

Transistor as a switch:

- **OFF:** input low, transistor in cut-off, $I_C \approx 0$, $V_{CE} \approx V_{CC}$.
- **ON:** input high, transistor in saturation, $V_{CE} \approx 0.2$ V; load gets almost full V_{CC}.
- Design: I_B must exceed I_C/β (give 2 to 5 times overdrive); $R_B = (V_{in} - 0.7)/I_B$.
- **Relay driver:** a microcontroller pin drives the base of an NPN transistor through R_B; the relay coil is in the collector; a **flyback diode** across the coil absorbs the voltage spike when the coil switches off.

MOSFET:

- **n-MOSFET (enhancement):** for $V_{GS} < V_T$, no channel (off). For $V_{GS} > V_T$, an inversion layer forms under the gate and current flows.
- In saturation, $I_D = k(V_{GS} - V_T)^2$.
- **MOSFET as a switch:** the gate takes almost no current, so a logic-level MOSFET is driven straight from a microcontroller pin. Low $R_{DS(on)}$ means little loss. Used for PWM motor and LED control.

BJT vs MOSFET:

Feature	BJT	MOSFET
Control	Current (base)	Voltage (gate)
Input resistance	Low to medium	Very high
Carriers	Bipolar	Unipolar
Switching speed	Moderate	Fast
Typical use	Small amplifiers, drivers	Digital ICs (CMOS), power switching

Power amplifiers: a **class A** amplifier conducts for the full 360° of the input, giving low distortion but a maximum efficiency of only 25% (resistive load) or 50% (transformer coupled). Classes B and AB conduct for about half the cycle and are more efficient.

🧮 Solved examples

Example 1 (BJT currents). A transistor has $\beta = 100$ and $I_B = 20$ µA. Find I_C, I_E and α.

- $I_C = 100 \times 20$ µA = **2 mA**; $I_E = 2.02$ mA; $\alpha = 100/101$ = **0.990**.

Example 2 (fixed bias Q-point). $V_{CC} = 12$ V, $R_B = 470$ kΩ, $R_C = 2.2$ kΩ, $\beta = 100$, $V_{BE} = 0.7$ V.

- $I_B = (12 - 0.7)/470$ kΩ = **24.0 µA**.
- $I_C = 100 \times 24.0$ µA = **2.40 mA**.

- $V_{CE} = 12 - 2.40 \times 2.2 =$ **6.71 V**. Since V_{CE} is well above 0.2 V and below V_{CC}, the transistor is in the active region.

Example 3 (relay driver). A 12 V relay coil draws 60 mA. The transistor has $\beta(\text{min}) = 50$ and is driven from a 5 V GPIO pin. Find R_B with 2 times overdrive.

- $I_B(\text{min}) = 60/50 = 1.2$ mA; with overdrive $I_B = 2.4$ mA.
- $R_B = (5 - 0.7)/2.4$ mA $= 1.79$ kΩ; use **1.8 kΩ** (gives 2.39 mA).

Example 4 (MOSFET). An n-MOSFET has $k = 0.5$ mA/V² and $V_T = 2$ V. At $V_{GS} = 4$ V, $I_D = 0.5 \times (4 - 2)^2 =$ **2 mA**.

Example 5 (PWM). A 5 V PWM signal with 25% duty cycle gives an average voltage of $0.25 \times 5 =$ **1.25 V**.

Key points to remember

- $I_E = I_B + I_C$; $\beta = I_C/I_B$; $\alpha = \beta/(\beta + 1)$.
- Active: BE forward, BC reverse. Saturation: both forward. Cut-off: both reverse.
- Fixed bias: $I_B = (V_{CC} - 0.7)/R_B$; $V_{CE} = V_{CC} - I_C R_C$.
- Switch: $R_B = (V_{in} - 0.7)/I_B$; always use a flyback diode with a relay.
- MOSFET: voltage controlled; $I_D = k(V_{GS} - V_T)^2$; class A efficiency 25% max.

⭐ Important questions

1. Explain the construction and working of an NPN transistor. Define α and β and derive the relation between them.
2. Explain the input and output characteristics of a CE configuration and mark the three regions.
3. Explain a CE amplifier and fixed bias with a numerical. Compare CB, CE and CC.
4. Explain the transistor as a switch. Design a relay driver circuit for a microcontroller.
5. Explain the construction and working of an n-channel enhancement MOSFET. Compare BJT and MOSFET. What is PWM?

UNIT 3 Operational Amplifiers

✍️ 2-mark definitions

- **Operational amplifier (op-amp):** a high-gain DC-coupled differential amplifier IC with two inputs (inverting −, non-inverting +) and one output. V_o = A_OL(V₊ − V₋).
- **Ideal op-amp:** infinite open-loop gain, infinite input resistance, zero output resistance, infinite bandwidth, infinite CMRR, zero offset and infinite slew rate.
- **Virtual short (virtual ground):** with negative feedback, V₊ ≈ V₋ because the gain is so high. If V₊ is grounded, V₋ is at 0 V without being connected to ground.
- **CMRR (common-mode rejection ratio):** the ratio of differential gain to common-mode gain, A_d/A_cm, usually in dB. High CMRR means noise common to both inputs is rejected.
- **Slew rate:** the maximum rate of change of the output voltage, in V/µs (0.5 V/µs for the 741).
- **Input offset voltage:** the small DC voltage that must be applied between inputs to make the output zero.
- **Comparator:** an op-amp used without feedback; the output swings to +V_sat or −V_sat depending on which input is larger.

💡 Core concepts

Practical 741 vs ideal:

Parameter	Ideal	741 (typical)
Open-loop gain	∞	2×10^5
Input resistance	∞	2 MΩ
Output resistance	0	75 Ω
Bandwidth	∞	1 MHz (unity gain)
CMRR	∞	90 dB
Slew rate	∞	0.5 V/µs

- Output cannot exceed the supply. With ±15 V supplies the saturation voltage is about ±13.5 V.

Standard circuits (derive using virtual short and zero input current):

1. **Inverting amplifier:** input through R_1 to the − input, R_f from output to − input, + input grounded. V_o = −(R_f/R_1)V_in.
2. **Non-inverting amplifier:** input to + input; R_1 from − input to ground and R_f from output to − input. V_o = (1 + R_f/R_1)V_in.
3. **Voltage follower (buffer):** output tied to − input, input on +. V_o = V_in, gain 1, very high input resistance. Used to stop a sensor being loaded.
4. **Summing amplifier (inverting):** V_o = −R_f(V_1/R_1 + V_2/R_2 + V_3/R_3). With equal resistors, V_o = −(V_1 + V_2 + V_3). Used as an audio mixer and a simple DAC.

5. **Difference amplifier:** with all resistors in ratio R_f/R_1, $V_o = (R_f/R_1)(V_2 - V_1)$.
6. **Comparator:** $V_o = +V_{sat}$ if $V_+ > V_-$, else $-V_{sat}$.

Derivation of inverting gain (write this in the exam):

- Because of the virtual ground, $V_- = 0$.
- Current through R_1: $I = V_{in}/R_1$. No current enters the op-amp, so the same current flows through R_f.
- $V_o = 0 - I R_f = -(R_f/R_1)V_{in}$.

Applications:

- **Automatic night lamp:** an LDR and a fixed resistor form a voltage divider into a comparator. In darkness the LDR resistance rises, the divider voltage crosses the reference set by a potentiometer, the comparator output goes high and switches on the lamp through a transistor or relay.
- **ECG amplifier:** heart signals are about 1 mV with large common-mode mains interference. A **differential (instrumentation) amplifier** with high CMRR amplifies only the difference between two electrodes and rejects the common 50 Hz noise.

🧮 Solved examples

Example 1 (inverting). $R_1 = 10\ k\Omega$, $R_f = 100\ k\Omega$, $V_{in} = 0.5$ V. Gain $= -100/10 =$ **−10**; $V_o =$ **−5 V**.

Example 2 (non-inverting). Same resistors: gain $= 1 + 100/10 =$ **11**; for 0.5 V in, $V_o =$ **5.5 V**.

Example 3 (summing). $R_f = R_1 = R_2 = R_3 = 10\ k\Omega$; inputs 1 V, 2 V, 3 V. $V_o = -(1 + 2 + 3) =$ **−6 V**.

Example 4 (saturation). An inverting amplifier of gain −20 is fed 1 V from ±15 V supplies. The ideal output would be −20 V, but the op-amp saturates at about **−13.5 V**.

Example 5 (difference amplifier). $R_f/R_1 = 10$, $V_2 = 2.1$ V, $V_1 = 2.0$ V. $V_o = 10 \times 0.1 =$ **1 V**.

Key points to remember

- Ideal: $A = \infty$, $R_{in} = \infty$, $R_{out} = 0$; virtual short $V_+ = V_-$.
- Inverting $-R_f/R_1$; non-inverting $1 + R_f/R_1$; follower 1.
- Summing $-R_f \Sigma V_i/R_i$; difference $(R_f/R_1)(V_2 - V_1)$.
- Comparator: no feedback, output $\pm V_{sat}$.
- 741: gain 2×10^5, R_{in} 2 MΩ, slew rate 0.5 V/µs, CMRR 90 dB.

⭐ Important questions

1. List the characteristics of an ideal op-amp and compare with a practical 741.
2. Derive the gain of inverting and non-inverting amplifiers. What is a virtual ground?
3. Explain the summing amplifier and the voltage follower with applications.
4. Explain the op-amp comparator and design an automatic night lamp.
5. Explain a differential amplifier and CMRR. Why is it used in an ECG amplifier?

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UNIT 4 Digital Electronics

✍️ 2-mark definitions

- **Number system:** a way of writing numbers with a base (radix): binary (2), octal (8), decimal (10), hexadecimal (16).
- **Logic levels:** voltage ranges read as 0 (LOW) and 1 (HIGH). In 5 V TTL, LOW is below 0.8 V and HIGH is above 2 V at the input; 3.3 V CMOS logic uses lower thresholds.
- **Universal gates:** NAND and NOR, because any Boolean function can be built using only one of them.
- **De Morgan's theorems:** (A + B)' = A'·B' and (A·B)' = A' + B'.
- **K-map (Karnaugh map):** a grid of 2^n cells in Gray-code order used to simplify a Boolean expression by grouping adjacent 1s in groups of 1, 2, 4, 8.
- **Combinational circuit:** output depends only on present inputs (adder, MUX). **Sequential circuit:** output depends on present inputs and past state (flip-flops, counters).
- **Multiplexer (MUX):** a data selector with 2^n inputs, n select lines and one output.
- **Flip-flop:** a 1-bit memory element with two stable states.

💡 Core concepts

Conversions:

- **Decimal to binary:** divide the integer part by 2 and read remainders bottom up; multiply the fraction by 2 and read the integer parts top down.
- **Binary to hex/octal:** group bits in fours (hex) or threes (octal) from the binary point.
- **2's complement:** invert all bits and add 1. Used to represent negative numbers and to subtract by adding.

Basic gates:

Gate	Expression	Output is 1 when
AND	A·B	All inputs are 1
OR	A + B	Any input is 1
NOT	A'	Input is 0
NAND	(A·B)'	Any input is 0
NOR	(A + B)'	All inputs are 0
XOR	$A \oplus B = A'B + AB'$	Inputs differ
XNOR	$(A \oplus B)'$	Inputs are equal

Boolean laws: identity (A + 0 = A, A·1 = A), null (A + 1 = 1, A·0 = 0), idempotent (A + A = A), complement (A + A' = 1, A·A' = 0), involution ((A')' = A), commutative, associative, distributive (A + BC = (A + B)(A + C)), absorption (A + AB = A, A(A + B) = A), and consensus.

K-map rules:

1. Draw the map with Gray-code labels (00, 01, 11, 10).
2. Put 1s for each minterm.
3. Make the largest possible groups of 1, 2, 4, 8 adjacent cells (edges wrap round; corners are adjacent in a 4-variable map).
4. Every 1 must be covered; overlap is allowed.
5. Each group gives a product term of the variables that do not change. Don't-care cells (X) may be used as 1s to make bigger groups.

Adders:

- **Half adder:** S = A ⊕ B; C = AB. (Two inputs, no carry in.)
- **Full adder:** S = A ⊕ B ⊕ C_in; C_out = AB + C_in(A ⊕ B). It can be built from two half adders and an OR gate.

Multiplexer (4:1): $Y = S_1'S_0'I_0 + S_1'S_0I_1 + S_1S_0'I_2 + S_1S_0I_3$.

Flip-flops (concept):

- **SR latch (NOR):** S = 1 sets Q = 1, R = 1 resets Q = 0, S = R = 0 holds, S = R = 1 is invalid.
- **D flip-flop:** on the clock edge, Q takes the value of D ($Q^+ = D$). It removes the invalid state and is the basic storage cell of registers.

S	R	Q^+ (SR)
0	0	Q (hold)
0	1	0 (reset)
1	0	1 (set)
1	1	Invalid

🧮 Solved examples

Example 1 (conversions).

- $(45.625)_{10}$: 45 = 101101 and 0.625 → 1.25 (1), 0.5 (0), 1.0 (1), so $\mathbf{(101101.101)_2}$.
- $(1AF)_{16} = 1 \times 256 + 10 \times 16 + 15 = \mathbf{(431)_{10}}$.
- −18 in 8-bit 2's complement: +18 = 00010010; invert 11101101; add 1 → **11101110**.

Example 2 (K-map, 3 variables). Simplify F(A, B, C) = Σm(1, 3, 5, 6, 7).

- The four cells with C = 1 (m1, m3, m5, m7) form a quad: **C**.
- m6 and m7 form a pair: **AB**.
- **F = C + AB**.

Example 3 (K-map, 4 variables). Simplify F(A, B, C, D) = Σm(0, 1, 2, 5, 8, 9, 10).

- Quad m0, m1, m8, m9 (B = 0, C = 0): **B'C'**.
- Quad of the four corners m0, m2, m8, m10 (B = 0, D = 0): **B'D'**.
- Pair m1, m5 (A = 0, C = 0, D = 1): **A'C'D**.
- **F = B'C' + B'D' + A'C'D**.

Example 4 (Boolean simplification). Simplify (AB + A'B)'. Inside the bracket, AB + A'B = B(A + A') = B, so the expression is **B'**. Also, by De Morgan, (A'B')' = A + B.

Key points to remember

- Conversions both ways; 2's complement = invert + 1.
- NAND and NOR are universal; De Morgan: break the bar, change the sign.
- K-map: groups of powers of 2, largest first, edges wrap.
- Half adder S = A $\oplus$ B, C = AB; full adder S = A $\oplus$ B $\oplus$ C_in, C_out = AB + C_in(A $\oplus$ B).
- SR: S = R = 1 invalid; D flip-flop Q^+ = D.

⭐ Important questions

1. Convert numbers between decimal, binary, octal and hex; find the 2's complement and subtract.
2. State and prove De Morgan's theorems. Show NAND and NOR are universal gates.
3. Simplify a 4-variable function using a K-map (with don't cares) and draw the logic circuit.
4. Design a half adder and a full adder; implement a full adder using half adders.
5. Explain a 4:1 multiplexer. Explain the SR latch and D flip-flop with truth tables.

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UNIT 5 Electronic Systems and Interfacing

✍️ 2-mark definitions

- **Microcontroller:** a single chip with a CPU, RAM, flash memory and peripherals (GPIO, timers, ADC, UART, SPI, I2C). Examples: ATmega328 (Arduino Uno), ESP32, STM32.
- **Microprocessor vs microcontroller:** a microprocessor is only the CPU and needs external memory and I/O (used in PCs); a microcontroller has everything on one chip (used in embedded systems).
- **GPIO:** general-purpose input/output pins that software can set as input (read a switch) or output (drive an LED).
- **ADC (analog-to-digital converter):** converts an analog voltage into an n-bit number. **Resolution** = $V_{ref}/2^n$.
- **Sensor:** a device that converts a physical quantity (temperature, light, pressure) into an electrical signal.
- **Baud rate:** the number of signal symbols per second on a serial line (for UART, bits per second).
- **LDO (low-dropout regulator):** a linear regulator that works with a very small difference between input and output (0.2 V to 1 V).
- **DC-DC (switching) converter:** converts one DC voltage to another by switching an inductor at high frequency. A **buck** steps down, a **boost** steps up. Efficiency is about 85% to 95%.

💡 Core concepts

ADC types:

- **Digital ramp (counter type):** a counter drives a DAC; when the DAC output exceeds V_{in}, a comparator stops the counter. Simple but slow: up to 2^n clock cycles per conversion.
- **Flash (parallel):** $2^n - 1$ comparators compare V_{in} with reference levels at the same time, and an encoder gives the code. Fastest, but needs many comparators (255 for 8 bits).
- **Successive approximation (SAR):** tries bits from MSB to LSB, n clock cycles. Used in most microcontrollers.

Serial communication:

Feature	UART	SPI	I2C
Wires	2 (TX, RX) + ground	4 (MOSI, MISO, SCK, CS)	2 (SDA, SCL)
Clock	None (asynchronous, agreed baud)	Shared (synchronous)	Shared (synchronous)
Devices	Point to point	One master, many slaves (one CS each)	Many devices by 7-bit address
Duplex	Full	Full	Half

Feature	UART	SPI	I2C
Speed	Up to about 1 Mbps	Tens of Mbps	100 kbps, 400 kbps, 3.4 Mbps
Typical use	GPS, Bluetooth modules, PC debug	SD card, display, fast ADC	Sensors, RTC, EEPROM

- **UART frame (8N1):** start bit, 8 data bits, no parity, 1 stop bit, so 10 bits per byte.

Sensor signal chain: sensor → signal conditioning (amplify with an op-amp, filter noise) → ADC → microcontroller (process) → output (display, actuator, wireless).

Wireless options:

Technology	Range	Data rate	Power	Use
Bluetooth / BLE	About 10 m	1 to 2 Mbps	Very low (BLE)	Wearables, phones
Wi-Fi	50 m indoors	Tens to hundreds of Mbps	High	Home IoT, internet
Zigbee	10 to 100 m (mesh)	250 kbps	Very low	Smart home, sensor networks

Power management:

- **LDO:** simple, no noise, but wastes $(V_{in} - V_{out}) \times I$ as heat; efficiency ≈ V_{out}/V_{in}.
- **Buck converter:** ideal duty cycle $D = V_{out}/V_{in}$; high efficiency; switching noise needs filtering.
- Battery life (hours) ≈ capacity (mAh)/average current (mA).

System examples (draw a block diagram each time):

1. **IoT weather station:** temperature and humidity sensor (DHT22 or BME280 on I2C), rain sensor and light sensor (ADC) → ESP32 → Wi-Fi to a cloud dashboard; powered by a solar panel, battery and buck or LDO regulator.
2. **Wearable fitness tracker:** accelerometer (steps), optical heart-rate sensor → low-power microcontroller → BLE to the phone; small Li-ion cell with a charger IC and LDO.
3. **EV battery management system (BMS):** measures the voltage of every cell, the pack current and temperatures; balances cells; protects against over-charge, over-discharge, over-current and over-temperature; estimates state of charge; reports on the vehicle's CAN bus.

🧮 Solved examples

Example 1 (ADC resolution). A 10-bit ADC with $V_{ref} = 3.3$ V. Resolution = 3.3/1024 = **3.22 mV**. An input of 1.65 V gives the code 1.65/0.00322 = **512**.

Example 2 (ADC types). For 8 bits, a flash ADC needs $2^8 - 1$ = **255 comparators**. A digital ramp ADC with a 1 MHz clock can take up to 2^8 = 256 cycles = **256 µs** in the worst case; a SAR ADC takes 8 cycles = **8 µs**.

Example 3 (UART). At 9600 baud with 8N1 framing, each byte takes 10 bits, so the data rate is 9600/10 = **960 bytes per second**.

Example 4 (LDO vs buck). 3.3 V at 200 mA is taken from 5 V.

- LDO: loss = (5 − 3.3) × 0.2 = **0.34 W**; efficiency = 3.3/5 = **66%**.
- Buck: ideal duty cycle D = 3.3/5 = **0.66**; real efficiency about 90%.

Example 5 (battery life). A 2000 mAh cell feeding an average of 50 mA lasts about 2000/50 = **40 hours**.

Key points to remember

- Microcontroller = CPU + memory + peripherals on one chip.
- ADC resolution $V_{ref}/2^n$; flash needs $2^n - 1$ comparators; ramp up to 2^n clocks.
- UART 2 wires, asynchronous; SPI 4 wires, fastest; I2C 2 wires, addressed.
- PWM average = D × V; UART 8N1 = 10 bits per byte.
- LDO efficiency ≈ V_{out}/V_{in}; buck D = V_{out}/V_{in}.

⭐ Important questions

1. Draw the block diagram of a microcontroller and explain GPIO, timers, ADC and PWM. Compare microprocessor and microcontroller.
2. Explain the digital ramp and flash ADCs. Find the resolution of an n-bit ADC.
3. Compare UART, SPI and I2C. Draw the UART frame.
4. Explain the sensor signal chain. Compare Bluetooth, Wi-Fi and Zigbee.
5. Design (block diagram) an IoT weather station, a wearable device or an EV BMS, including power management (LDO and DC-DC).

Last-hour checklist

☐ pn junction bias and V-I; HWR 3.82 V, bridge 7.64 V; rectifier table (γ 1.21 vs 0.482; η 40.6% vs 81.2%; PIV); ripple 1 V; Zener (31.4, 10, 21.4 mA, 0.109 W); LED 300 Ω; 5 V supply chain.

☐ BJT α, β (2 mA, 0.990); regions; fixed bias (24 µA, 2.40 mA, 6.71 V); CB/CE/CC table; relay driver 1.8 kΩ with flyback diode; MOSFET 2 mA; PWM 1.25 V; class A 25%.

☐ Ideal op-amp and 741 table; virtual ground; inverting −10, non-inverting 11, summing −6 V, saturation 13.5 V, difference 1 V; night lamp; ECG and CMRR.

☐ 101101.101; 431; 11101110; gates table; De Morgan; K-maps (C + AB; B'C' + B'D' + A'C'D); adders; 4:1 MUX; SR and D flip-flop.

☐ Microcontroller blocks; ADC 3.22 mV, 512; 255 comparators, 256 µs; UART/SPI/I2C table; 960 bytes/s; LDO 0.34 W, 66%; buck 0.66; 40 h; three system block diagrams.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Electronic Devices and Circuit Theory	Robert L. Boylestad, Louis Nashelsky	Pearson	WorldCat
2	Electronics Fundamentals and Applications	D. Chattopadhyay, P. C. Rakshit	New Age International	WorldCat
3	Digital Design	M. Morris Mano	Pearson	WorldCat
4	Op-Amps and Linear Integrated Circuits	Ramakant A. Gayakwad	Pearson	WorldCat

Syllabus source: AKTU B.Tech 1st year CSE stream course structure and syllabus, 2026-27 (official PDF)

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